5.17 P.20

$L(t) = \sqrt{13t^2 - 42t + 34}$ pour $0 \leq t \leq 3$

a) $L(2) = \sqrt{2}$ mètre

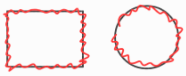
b) graphique T.I

c) les 2 objets se rapproche de $[0 ; 1{,}62[$
les 2 objets s'éloigne de $]1.62 ; 3]$

d) à 1,62 seconde la distance est proche de 0

5.16 P.20

fil de fer de 100 cm

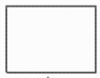
a) 

30 cm 70 cm

$r = 11{,}14$

$\frac{30}{4} = 7{,}5 \rightarrow c$

$A_{total} = c^2 + \pi \cdot r^2 = 446{,}18\ cm^2$

b) 

$\frac{x}{4}$

$\frac{x}{4}$

$A_{carré} = \left(\frac{x}{4}\right)^2 \Rightarrow \frac{x^2}{16}$

$100 = x + 2\pi r \qquad r = \frac{100 - x}{2\pi}$

$A_{cercle} = \pi r^2 \rightarrow \pi \cdot \left(\frac{100-x}{2\pi}\right)^2 \rightarrow \cancel{\pi} \cdot \frac{(100-x)^2}{4\pi^{\cancel{2}}}$

$A_{total} = \frac{(x^2)\cdot\pi}{(16)\cdot\pi} + \frac{((100-x)^2)\cdot 4}{(4\pi)\cdot 4} \Rightarrow \frac{(x^2)\cdot\pi + (100-x)^2 \cdot 4}{16\pi} \Rightarrow \frac{(x^2)\cdot\pi + (10000 - 100x - 100x + x^2)\cdot 4}{16\pi}$

$= \frac{(x^2)\cdot\pi + 40\,000 - 800x + 4x^2}{16\pi} \Rightarrow \frac{(\pi + 4)\cdot x^2 + 40000 - 800x}{16\pi}$

$(100-x)^2 = (100-x)\cdot(100-x) \Rightarrow 10000 - 100x - 100x + x^2$

5.24 P.32

$(f \circ g)(x) \qquad (g \circ f)(x)$

e) $f(x) = \sqrt{3-x} \qquad g(x) = x^2 + 3$

$(f \circ g)(x) = f(g(x)) = f(x^2+3) = \sqrt{3-(x^2+3)}$

$= \sqrt{-x^2}$ = non-def

$(g \circ f)(x) = g(f(x)) = g(\sqrt{3-x}) = (\sqrt{3-x})^2 + 3$

$= 3 - x + 3$

$= 6 - x$

f) $f(x) = 2x + 1 \qquad g(x) = \frac{5+x}{5-x}$

$f(g(x)) = f\left(\frac{5+x}{3-x}\right) = 2\left(\frac{5+x}{5-x}\right) + 1$

$g(f(x)) = g(2x+1) = \frac{5 + (2x+1)}{5 - (2x+1)} \Rightarrow \frac{2x+6}{2x-4} \Rightarrow \frac{\cancel{2}(x+3)}{\cancel{2}(x-2)} = \frac{x+3}{x-2}$

5.25

d) $g(x) = 3x$

$f(x) = \frac{1}{x \quad +2}$

$h(x) = (f \circ g)(x) = f(g(x)) = f(3x) = \frac{1}{3x+2}$

5.28 P.33

a) $f(x) = x^3 - 5$

$y = x^3 - 5 \Rightarrow y + 5 = x^3 \Rightarrow \sqrt[3]{y+5} = x \qquad f^{-1}(x) = \sqrt[3]{x+5}$

$f \circ f^{-1}$

$f(\sqrt[3]{x+5}) \Rightarrow (\sqrt[3]{x+5})^3 - 5 \Rightarrow x + 5 - 5 \Rightarrow x$

$f^{-1} \circ f$

$f^{-1}(x^3 - 5) \Rightarrow \sqrt[3]{(x^3-5)+5} \Rightarrow \sqrt[3]{x^3} \Rightarrow x$

6.6 P.47

a) $C(x) = 200x + 20\,000$

b) $R(x) = 500x$

c) $C(x) = R(x) \Rightarrow \underset{-200x}{200x} + 20000 = \underset{-200x}{500x} \Rightarrow \frac{20000}{300} = \frac{300x}{300} \Rightarrow \frac{200}{3} = x \longrightarrow 66.66667$ pièces

d) Pt intersection est $(66,7\,;\,33\,300)$

e) $P(x) = R(x) - C(x) \Rightarrow P(x) = 300x - 20\,000$

f) vérifié

g) $P(x) = 300x - 20000$

$y = 300x - 20000 \rightarrow \frac{y + 20\,000}{300} = \frac{300x}{300} \longrightarrow \frac{y + 20\,000}{300} = x$

switch

$P^{-1}(x) = \frac{x + 20\,000}{300}$

$P^{-1}(50\,000) = \frac{50\,000 + 20000}{300} = \frac{700}{3}$ pour un profit de 50K il faut faire $\frac{700}{3}$ pièces